

HIGHER DIPLOMA IN INFORMATION TECHNOLOGY

Web Application Development

Semester - 04

Repeat Assignment

Student name: G.W. Chalitha Vinodya

Student ID: OC/24/HDIT/B2/4819

1. Introduction

Orion College currently manages its student hostel rooms manually using physical paper records and basic spreadsheets across its campus facilities. This conventional process has led to major operational bottlenecks, including critical difficulties in allocating rooms, frequent loss of reservation records, incorrect payment tracking, duplicate room bookings, and severe difficulties in monitoring real-time room availability for administrative staff.

To resolve these systemic inefficiencies, I have designed and developed a fully functional, web-based Online Hostel & Room Reservation Management System. This system transitions Orion College from a vulnerable, paper-heavy approach into a centralized, secure, dynamic, and data-driven infrastructure. The application provides an integrated digital workspace that automates administrative tasks while giving students a transparent, self-service web portal to manage their accommodation requirements.

2. Objectives

The primary objective of my project is to engineer a secure web application that eliminates manual workflows and establishes clean data integrity. My specific project objectives are:

- **To Centralize Data Architecture:** Replace physical logbooks and isolated spreadsheets with a structured MySQL relational database that securely houses all user credentials, room allocations, bookings, and financial logs.
- **To Eliminate Duplicate Bookings:** Program an automated room availability checker that alters a room's status instantly upon reservation approval, preventing dual-allocation conflicts.
- **To Implement Accurate Payment Tracking:** Deploy a transactional ledger system to log and map payments directly against approved booking references, establishing audit transparency.
- **To Streamline Room Allocation and Monitoring:** Create a dynamic analytics dashboard for administration users to monitor live vacancy counts instantly.
- **To Deliver a Student Self-Service Portal:** Enable students to browse live room vacancies, submit booking requests, and view their personal payment history autonomously.

3. System Analysis

During the system analysis phase, I evaluated the core requirements of Orion College to map out the application's boundaries.

Functional Requirements

- **Role-Based Authentication:** A secure login gate that identifies whether a user is an Admin or a Student, restricting access to features based on their role.
- **Room Management (CRUD):** Administrative privileges to Create new room records, Read/View current configurations, Update pricing or capacity, and Delete records securely.
- **Reservation Engine:** A streamlined application workspace for students to choose available rooms and input required check-in/check-out schedules.

- **Payment Log Tracking:** An interface that captures financial transaction values and links them to approved bookings.

Non-Functional Requirements

- **Security:** Defensive coding structures using Prepared Statements to block SQL Injection threats and implementation of input sanitization.
- **Usability:** An intuitive user interface with crisp typography, a consistent color palette, and straightforward navigation sidebars.
- **Data Consistency:** Database foreign key constraints that prevent orphan data entries (e.g., ensuring payments cannot exist without a valid reservation reference).

4. System Design

My system layout is divided into a presentation layer (HTML5, CSS3, JavaScript), a server side processing layer (PHP 8), and a data persistence layer (MySQL).

Database Design

I designed a relational schema containing four highly normalized tables with explicit relationships:

1. **users Table:** Contains user profile assets. Attributes include id (PK), username, email (Unique), password (stored as a secure hash), and role (Admin/Student).
2. **rooms Table:** Tracks room assets. Attributes include id (PK), room_number (Unique), room_type, capacity, price, status (Available/Reserved), and image (file path string).
3. **reservations Table:** Binds student profiles to physical rooms. Attributes include id (PK), student_name, student_id, room_id (FK linking to rooms.id), check_in, check_out, and status (Pending/Approved/Rejected).
4. **payments Table:** Logs financial collections. Attributes include id (PK), reservation_id (FK linking to reservations.id), amount, payment_date, and payment_status.

```

29
30
31 Database: 'hotel_management_db'
32
33 .....
34
35 -- Table structure for table 'payments'
36
37 CREATE TABLE 'payments' (
38   'id' int(11) NOT NULL,
39   'reservation_id' int(11) NOT NULL,
40   'amount' decimal(10,2) NOT NULL,
41   'payment_date' timestamp NOT NULL DEFAULT current_timestamp(),
42   'payment_status' enum('Pending','Paid')
43   ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
44
45 -- Dumping data for table 'payments'
46
47 INSERT INTO 'payments' ('id','reservation_id','amount','payment_date','payment_status') VALUES
48 (1,1,5000.00,'2020-05-12 00:00:13','Paid');
49
50 .....
51
52 -- Table structure for table 'reservations'
53
54 CREATE TABLE 'reservations' (
55   'id' int(11) NOT NULL,
56   'student_name' varchar(100) NOT NULL,
57   'student_id' varchar(50) NOT NULL,
58   'room_id' int(11) NOT NULL,
59   'reservation_date' timestamp NOT NULL DEFAULT current_timestamp(),
60   'check_in' date NOT NULL,
61   'check_out' date NOT NULL,
62   'status' varchar(50) DEFAULT 'Pending'
63   ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
64
65 -- Dumping data for table 'reservations'
66
67 INSERT INTO 'reservations' ('id','student_name','student_id','room_id','reservation_date','check_in','check_out','status') VALUES
68 (1,'James','002014889','1','2020-05-12 00:10:29','2020-06-01','2020-06-05','Approved');
  
```

5. Wireframes & 6. UI Designs

I drafted low-fidelity wireframes in Figma to outline the structural skeleton of the system before polishing them into high-fidelity user interface concepts. My design choices utilize a professional deep blue primary color (#1e3a8a) to establish institutional credibility, paired with soft light slate backdrops (#f4f6f9) to maximize text readability.

The layout incorporates a permanent left navigation sidebar for quick swapping between screens, and information fields are presented cleanly in responsive cards to draw focus to core statistical metrics.

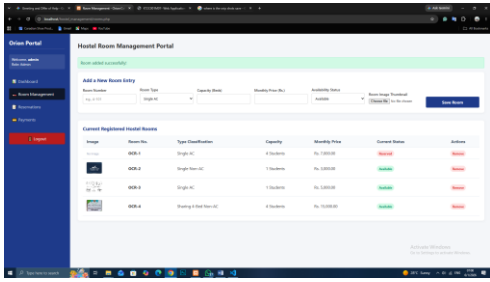
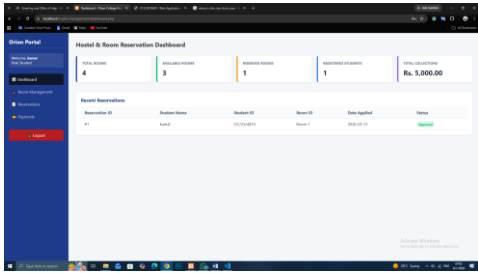
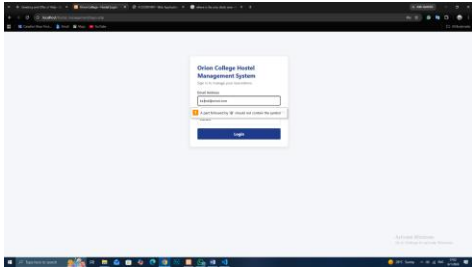
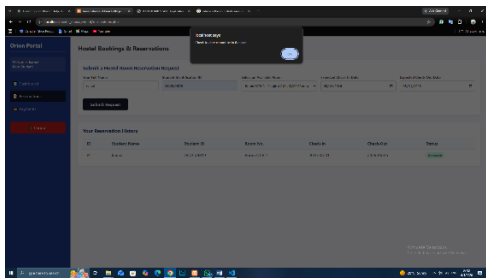
7. System Implementation

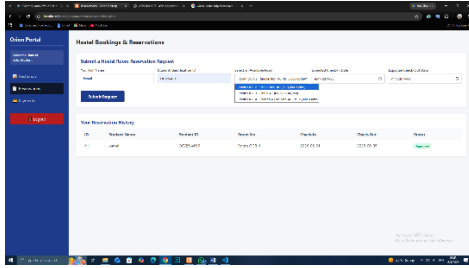
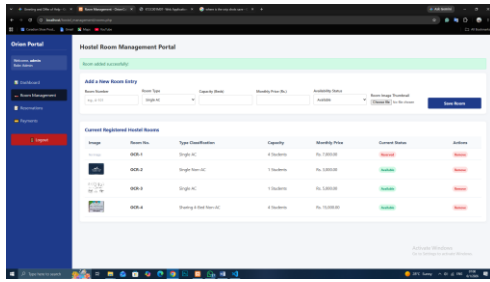
I implemented the system using a native open-source technology stack: **HTML5, CSS3, JavaScript (ES6), PHP 8, and MySQL.**

- **Security Controls:** To safeguard user data, I avoided storing plaintext passwords. Instead, I implemented secure cryptographic encryption using PHP's native `password_hash()` function. Furthermore, I built all database interactions using **Prepared Statements** (`$conn->prepare()`) to completely block potential SQL Injection attacks.
- **Advanced Features:** To satisfy the module's advanced development criteria, I implemented two specific advanced features:
 1. *Responsive Mobile-Friendly Design:* I utilized CSS Flexbox and media queries to ensure the entire layout adjusts seamlessly across mobile phones, tablets, and desktop views.
 2. *Image Upload Functionality:* I built an automated backend image processing engine inside `rooms.php` that permits administrators to upload real physical room preview images to the server directory, dynamically renaming them using UNIX timestamps to prevent filename duplication conflicts.

8. Testing

To guarantee application reliability and prove that all of Orion College's manual vulnerabilities have been programmatically resolved, I executed comprehensive functional tests based on the testing matrix below:

Test Case No	Scenario Description	Expected Output	Actual Output	Result
T01	Add a New Room (Admin)	Room record saves; image file transfers to server folder.		Pass
T02	Secure Portal Login	Authenticate valid inputs and initiate session parameters.		Pass
T03	Blank Fields Guard	Prevent login if required fields are omitted.		Pass
T04	Past Date Booking Attempt	Prevent student from picking a past date for check-in.		Pass

Test Case No	Scenario Description	Expected Output	Actual Output	Result
T05	Double Booking Prevention	Approved rooms must instantly vanish from vacancy dropdown selection.		Pass
T06	Upload room photos	Room photos must see in system		Pass

9. Conclusion

The Online Hostel & Room Reservation Management System designed for Orion College successfully addresses all the critical pain points identified in the manual workflow scenario. By implementing a secure, structured database environment coupled with automated business logic, the platform removes room allocation conflicts, eliminates data loss risks, prevents duplicate bookings, and ensures full payment transparency. The source code and relational database scripts are complete and ready for campus-wide network deployment.

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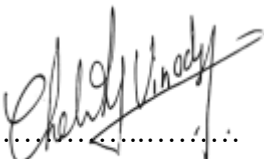
Competency Standards Code: K72C001M07

DECLARATION OF ACADEMIC HONESTY

STUDENT NAME:	G.W.Chalitha Vinodya
STUDENT REGISTRATION NUMBER:	4819

MODULE TITLE:	Web Application Development
TASK NUMBER:	
ASSIGNMENT/PROJECT TITLE:	

ASSIGNMENT TYPE:	
ISSUED DATE:	2026/05/15
DUE DATE:	2026/06/01
DATE SUBMITTED:	2026/06/01
ASSESSOR:	Ms. Hasara Liyanagamage
INTERNAL VERIFIER:	Mr. R. H. Dilruksh
EXTERNAL VERIFIER:	

Student signature: 

Date: ...01... / ...06.... / 2026

Grading System

Score (marks)	Grade	Grade Point Average (GPA)
80-100	A+	4.00
75-79	A	3.75
70-74	A-	3.50
65-69	B+	3.25
60-64	B	3.00
55-59	B-	2.75
50-54	C+	2.50
45-49	C	2.25
40-44	C-	2.00
30-39	D	0.00 (Fail)
0-29	E	0.00 (Fail)
Incomplete	I	-
No Credit	NC	-
Audit	AU	-
Withdrawn	W	-
Withdrawn-failing	WF	-
Repeat completed	RC	-

Task 01 Marks	Task 01 Marks	Task 01 Marks	Task 01 Marks	Total Marks

Total Score (<i>Average of the above/ Overall Grading</i>)	Grade Point	Final Grading

Assessor Comment:

Assessor Signature.....

Date: / /

Internal Verifier Comment:

Internal Verifier Signature.....

Date: / /

External Verifier Comment:

External Verifier Signature.....

Date: / /

Instructions to Students

Deliverable: Hard & Soft copy of word-processed report

Note:

- You Should include the **COVER PAGE** of this assignment including the **Declaration of academic Honesty** when you submit your final report.
- More work is required to achieve a pass. If you submit your assignments after the deadline set above your assignment will be adding -5 point per day provided it falls within the extended deadline.
- If submitted after the extended deadline, the assignment will not be accepted whereas you shall be asked to go for a **NEW** Assignment.
- Plagiarism is strictly prohibited and will lead for a "Fail" Grade

Instructions to write an assignment

1. All assignments should comprise of the standard **Front Cover** given. **No other front page will be accepted.**
2. All assignments should be bound with Black hard paper for back and clear polymer sheet as the front cover.
3. **Report writing guidelines:**
 - a. Every assignment should have an **Introduction** and **Conclusion**.
 - b. The Standard Automatic Table of Content should be generated.
 - c. All the figures, Table Diagrams, etc. should be numbered.
 - d. Main Heading Font: **Times New Roman – size 16 - Bold**
 - e. Sub Heading Font: **Times New Roman – size 14 - Bold**
 - f. Body Text Font: **Times New Roman – size 12**
 - g. Line Space: After 8pt, Before 0pt, Multiple at 1.15
 - h. Margins: **Top – 1" Bottom – 1" Left-1" Right-1"**
 - i. **Header:** Include the Module name on Right hand side.
 - j. **Footer:** Include the college name on left and Page number on right hand side.
 - k. All sections should have continuity and pages should be clearly labelled.
 - l. **References: APA6 or APA7** (Must generate Bibliography at the end of the assignment).

Declaration of Academic Honesty Instructions

1. You must submit a completed copy of this form every time you submit an assignment.
2. You must submit declaration electronically, either within your assignment, file, or as a separate file.
3. If you do not submit this declaration, your work will not be assessed.
4. If any student is found to have been dishonest, then the student may face disciplinary action as stated in the Orion College Plagiarism policy.

Plagiarism Warning for Students

Original Work: All assignments submitted for this course must be your own original work. This means that you should develop your solutions independently and avoid copying from any source, including other students, websites, or published materials, without proper citation.

Proper Citation: When referencing external sources, such as textbooks, academic papers, or online resources, it's essential to provide accurate citations in accordance with the appropriate citation style (e.g., APA6 or APA 7). This includes both direct quotations and paraphrased information.

Avoiding Code Plagiarism: In programming assignments, it's common for students to encounter existing code online. While it's acceptable to seek guidance and reference external resources, it's crucial that you understand and modify the code to suit the specific requirements of the assignment. Directly copying and submitting code without proper attribution is considered plagiarism.

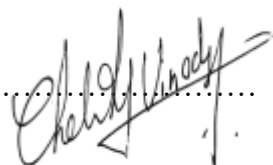
Collaboration Guidelines: Collaboration with fellow students may be allowed in certain instances, but it's essential to adhere to the collaboration policy outlined by the instructor. If collaboration is permitted, make sure to acknowledge the contributions of your collaborators and submit your own unique work.

Consequences of Plagiarism: Plagiarism is a serious academic offense that can result in severe consequences, including failing grades on assignments, failing the course, or even academic probation or expulsion from the university. It's not worth risking your academic and professional reputation for the sake of convenience.

Declaration

I confirm that this assignment is my own work & I have clearly indicated in my assignment if have any work that has been carried out collaboratively with another student.

Student signature:



Date: ...01... / ...06.... / ...2026...